

1 Product Measures

We work on the product space $(X, \mathcal{F}) \times (Y, \mathcal{G}) \cong (X \times Y, \mathcal{F} \otimes \mathcal{G})$ where

$$\mathcal{F} \otimes \mathcal{G} := \sigma(\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}).$$

We want to define a product measure $\mu \otimes \nu$ from $(X, \mathcal{F}, \mu) \times (Y, \mathcal{G}, \nu)$ where μ and ν are finite measures.

For $E \in \mathcal{F} \otimes \mathcal{G}$, define the x -section $E^x := \{y \in Y : (x, y) \in E\}$

Lemma 1.1. For every $E \in \mathcal{F} \otimes \mathcal{G}$ and every $x \in X$, $E^x \in \mathcal{G}$

Proof. Fix $x \in X$ and define $\mathcal{H}_x := \{E \in \mathcal{F} \otimes \mathcal{G} : E^x \in \mathcal{G}\}$. Then $\mathcal{H}_x$ is a σ -algebra, since

$$(E^c)^x = (E^x)^c, \quad \left(\bigcup_{n=1}^{\infty} E_n \right)^x = \bigcup_{n=1}^{\infty} E_n^x.$$

Moreover, for $A \in \mathcal{F}$, $B \in \mathcal{G}$,

$$(A \times B)^x = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A, \end{cases}$$

so $A \times B \in \mathcal{H}_x$. Hence

$$\mathcal{F} \otimes \mathcal{G} = \sigma(\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}) \subseteq \mathcal{H}_x.$$

Thus $E^x \in \mathcal{G}$

□

For $E \in \mathcal{F} \otimes \mathcal{G}$, define

$$f^E : X \rightarrow [0, \infty], \quad f^E(x) := \nu(E^x).$$

Lemma 1.2. For every $E \in \mathcal{F} \otimes \mathcal{G}$, $f^E \geq 0$ and f^E is $\mathcal{F}/\mathcal{B}([0, \infty])$ -measurable.

Proof. (A tempting approach.) Let

$$\mathcal{H} := \{E \in \mathcal{F} \otimes \mathcal{G} : f^E \text{ is measurable}\}.$$

One might try to show that $\mathcal{H}$ is a σ -algebra containing all measurable rectangles. Then, since

$$\mathcal{F} \otimes \mathcal{G} = \sigma(\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}),$$

the result would follow.

The difficulty is closure under arbitrary countable unions. Suppose $E_1, E_2, \dots \in \mathcal{H}$. We have

$$\left(\bigcup_{n=1}^{\infty} E_n \right)^x = \bigcup_{n=1}^{\infty} E_n^x,$$

but it is not immediate from the measurability of the functions f^{E_n} that the map $x \mapsto \nu(\bigcup_{n=1}^{\infty} E_n^x)$ is measurable.

One can try to disjointify the sets:

$$F_1 = E_1, \quad F_n = E_n \setminus \bigcup_{k=1}^{n-1} E_k.$$

Then $f^{\cup_{n=1}^{\infty} E_n} = \sum_{n=1}^{\infty} f^{E_n}$. However, this requires that

$$E, F \in \mathcal{H} \implies E \setminus F \in \mathcal{H}.$$

But

$$f^{E \setminus F}(x) = \nu(E^x \setminus F^x) = \nu(E^x) - \nu(E^x \cap F^x),$$

and the measurability of the second term is not yet known. Thus the direct σ -algebra argument leads to an unnecessary closure-under-differences and intersections problem.

(Correct approach using π - λ theorem.)

Instead, we show that $\mathcal{H}$ is a λ -system and that it contains the π -system of measurable rectangles.

Remark 1.3. The measurable rectangles form a π -system. If $A, C \in \mathcal{F}$ and $B, D \in \mathcal{G}$, then

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D).$$

Recall that a λ -system contains the whole space, is closed under complements, and is closed under countable disjoint unions.

Let $\mathcal{H} := \{E \in \mathcal{F} \otimes \mathcal{G} : f^E \text{ is measurable}\}$. We show that $\mathcal{H}$ is a λ -system.

1. $X \times Y \in \mathcal{H}$ since $f^{X \times Y}(x) = \nu(Y)$ which is measurable.
2. Suppose $E \in \mathcal{H}$. Then $(E^c)^x = (E^x)^c$ since ,

$$\begin{aligned} f^{E^c}(x) &= \nu((E^x)^c) \\ &= \nu(Y) - \nu(E^x) = \nu(Y) - f^E(x). \end{aligned}$$

Hence f^{E^c} is measurable, and therefore $E^c \in \mathcal{H}$.

3. Let $E_1, E_2, \dots \in \mathcal{H}$ be pairwise disjoint and put

$$E = \bigcup_{n=1}^{\infty} E_n \implies E^x = \bigcup_{n=1}^{\infty} E_n^x$$

where the E_n^x are pairwise disjoint. Thus

$$\begin{aligned} f^E(x) &= \nu\left(\bigcup_{n=1}^{\infty} E_n^x\right) \\ &= \sum_{n=1}^{\infty} \nu(E_n^x) = \sum_{n=1}^{\infty} f^{E_n}(x). \end{aligned}$$

Since each f^{E_n} is nonnegative and measurable, f^E is measurable. Hence $\bigcup_{n=1}^{\infty} E_n \in \mathcal{H}$.

Finally, for $A \in \mathcal{F}$, $B \in \mathcal{G}$, $f^{A \times B} = \nu(B)\mathbf{1}_A$ is measurable, so $A \times B \in \mathcal{H}$. Since the measurable rectangles form a π -system, the π - λ theorem yields

$$\mathcal{F} \otimes \mathcal{G} = \sigma(\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}) \subseteq \mathcal{H}.$$

Thus $\mathcal{H} = \mathcal{F} \otimes \mathcal{G}$. Therefore is measurable for every $E \in \mathcal{F} \otimes \mathcal{G}$. □

By the lemma, f^E is nonnegative and measurable for every $E \in \mathcal{F} \otimes \mathcal{G}$, so the integral $\int_X f^E d\mu$ is well-defined in $[0, \infty]$.

Now define $\pi : \mathcal{F} \otimes \mathcal{G} \rightarrow [0, \infty]$ by

$$\pi(E) := \int_X f^E d\mu.$$

Proposition 1.4. For every $A \in \mathcal{F}$ and $B \in \mathcal{G}$, $\pi(A \times B) = \mu(A)\nu(B)$.

Proof. For a measurable rectangle $A \times B$,

$$(A \times B)^x = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A. \end{cases}$$

Hence $f^{A \times B}(x) = \nu(B)\mathbf{1}_A(x)$ and therefore

$$\begin{aligned} \pi(A \times B) &= \int_X f^{A \times B} d\mu \\ &= \int_X \nu(B)\mathbf{1}_A d\mu = \nu(B)\mu(A). \end{aligned}$$

□

Theorem 1.5. π is a measure on $(X \times Y, \mathcal{F} \otimes \mathcal{G})$. Moreover, for $A \in \mathcal{F}$, $B \in \mathcal{G}$,

$$\pi(A \times B) = \mu(A)\nu(B).$$

In particular,

$$\pi(X \times Y) = \mu(X)\nu(Y) < \infty,$$

so π is a finite measure.

Finally, if π' is any other measure on $(X \times Y, \mathcal{F} \otimes \mathcal{G})$ such that

$$\pi'(A \times B) = \mu(A)\nu(B) \quad \forall A \in \mathcal{F}, \forall B \in \mathcal{G},$$

then $\pi' = \pi$. Thus π is the *product measure*, denoted

$$\boxed{\mu \otimes \nu}.$$

Proof. From the definition of π ,

$$\pi(A \times B) = \int_X f^{A \times B} d\mu = \int_X \nu(B)\mathbf{1}_A d\mu.$$

Therefore $\pi(A \times B) = \nu(B)\mu(A)$. Furthermore,

$$\pi(X \times Y) = \mu(X)\nu(Y) < \infty.$$

Since

$$\pi(\emptyset) = \int_X f^\emptyset d\mu = 0,$$

it remains to prove countable additivity.

Let $E_1, E_2, \dots \in \mathcal{F} \otimes \mathcal{G}$ be pairwise disjoint. Then

$$\begin{aligned} \pi\left(\bigcup_{n=1}^{\infty} E_n\right) &= \int_X f^{\bigcup_{n=1}^{\infty} E_n} d\mu \\ &= \int_X \left(\sum_{n=1}^{\infty} f^{E_n}\right) d\mu \\ &= \lim_{N \rightarrow \infty} \int_X \sum_{n=1}^N f^{E_n} d\mu \quad (\text{MCT}) \\ &= \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_X f^{E_n} d\mu = \sum_{n=1}^{\infty} \pi(E_n). \end{aligned}$$

Hence π is a measure on $(X \times Y, \mathcal{F} \otimes \mathcal{G})$. Since the measurable rectangles form a π -system and

$$\pi(A \times B) = \mu(A)\nu(B),$$

the uniqueness result from the π - λ theorem gives uniqueness of π . □

Definition 1.6. The measure

$$\pi : \mathcal{F} \otimes \mathcal{G} \rightarrow [0, \infty], \quad E \mapsto \pi(E),$$

satisfying

$$\pi(A \times B) = \mu(A)\nu(B) \quad \forall A \in \mathcal{F}, B \in \mathcal{G},$$

is called the *product measure*, denoted by

$$\boxed{\mu \otimes \nu}.$$

Symmetric formulation.

Define the y -section

$$E_y := \{x \in X : (x, y) \in E\}$$

and

$$g_E : Y \rightarrow [0, \infty], \quad g_E(y) := \mu(E_y).$$

Interchanging the roles of $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$, the same construction gives a measure

$$\pi'(E) := \int_Y g_E \, d\nu.$$

For $A \in \mathcal{F}, B \in \mathcal{G}$,

$$\pi'(A \times B) = \mu(A)\nu(B) = \pi(A \times B).$$

By uniqueness of the product measure,

$$\pi' = \pi.$$

Hence

$$\boxed{\pi(E) = \int_X \nu(E^x) \, d\mu = \int_Y \mu(E_y) \, d\nu.}$$

The σ -finite case.

For the σ -finite case, suppose

$$\mu \text{ is } \sigma\text{-finite on } (X, \mathcal{F}).$$

Then there exist pairwise disjoint sets

$$A_m \in \mathcal{F}, \quad X = \bigcup_{m=1}^{\infty} A_m, \quad \mu(A_m) < \infty.$$

Similarly, if ν is σ -finite, there exist pairwise disjoint

$$B_n \in \mathcal{G}, \quad Y = \bigcup_{n=1}^{\infty} B_n, \quad \nu(B_n) < \infty.$$

Assume that μ and ν are σ -finite. Choose pairwise disjoint measurable sets

$$X = \bigcup_{m=1}^{\infty} A_m, \quad Y = \bigcup_{n=1}^{\infty} B_n,$$

such that

$$\mu(A_m) < \infty, \quad \nu(B_n) < \infty.$$

Define the restricted measures

$$\mu_m := \mu|_{A_m}, \quad \mu_m(A) = \mu(A \cap A_m) < \infty,$$

so μ_m is a finite measure. Similarly,

$$\nu_n := \nu|_{B_n}, \quad \nu_n(B) = \nu(B \cap B_n),$$

and ν_n is also a finite measure.

Since the product measure has already been constructed for finite measures, define

$$\pi_{m,n} := \mu_m \otimes \nu_n.$$

We then define

$$\pi := \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi_{m,n}.$$

Since a countable sum of measures is again a measure,

$$\pi(E) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi_{m,n}(E), \quad E \in \mathcal{F} \otimes \mathcal{G},$$

defines a measure on

$$(X \times Y, \mathcal{F} \otimes \mathcal{G}).$$

Since

$$X \times Y = \bigcup_{m=1}^{\infty} \bigcup_{n=1}^{\infty} (A_m \times B_n),$$

and the rectangles $A_m \times B_n$ are pairwise disjoint, π is σ -finite. Indeed,

$$\begin{aligned} \pi(A_{m_0} \times B_{n_0}) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi_{m,n}(A_{m_0} \times B_{n_0}) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \mu_m(A_{m_0}) \nu_n(B_{n_0}) \\ &= \mu(A_{m_0}) \nu(B_{n_0}) < \infty, \end{aligned}$$

because

$$\mu_m(A_{m_0}) = \begin{cases} \mu(A_{m_0}), & m = m_0, \\ 0, & m \neq m_0, \end{cases}$$

and similarly

$$\nu_n(B_{n_0}) = \begin{cases} \nu(B_{n_0}), & n = n_0, \\ 0, & n \neq n_0. \end{cases}$$

Now, for $A \in \mathcal{F}$, $B \in \mathcal{G}$,

$$\begin{aligned} \pi(A \times B) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi_{m,n}(A \times B) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \mu_m(A) \nu_n(B) \\ &= \left(\sum_{m=1}^{\infty} \mu_m(A) \right) \left(\sum_{n=1}^{\infty} \nu_n(B) \right) = \mu(A) \nu(B). \end{aligned}$$

Indeed, by the disjoint decompositions of X and Y ,

$$\begin{aligned} \sum_{m=1}^{\infty} \mu_m(A) &= \sum_{m=1}^{\infty} \mu(A \cap A_m) = \mu(A) \\ \sum_{n=1}^{\infty} \nu_n(B) &= \sum_{n=1}^{\infty} \nu(B \cap B_n) = \nu(B) \end{aligned}$$

Thus

$$\boxed{\pi(A \times B) = \mu(A) \nu(B)} \quad \forall A \in \mathcal{F}, B \in \mathcal{G}.$$

Theorem 1.7. Let π' be any σ -finite measure on $(X \times Y, \mathcal{F} \otimes \mathcal{G})$ such that

$$\pi'(A \times B) = \mu(A)\nu(B) \quad \forall A \in \mathcal{F}, B \in \mathcal{G}.$$

Then $\pi' = \pi$.

Proof. Fix m, n and define $R_{m,n} := A_m \times B_n$. Then

$$\pi(R_{m,n}) = \mu(A_m)\nu(B_n) < \infty, \quad \pi'(R_{m,n}) = \mu(A_m)\nu(B_n) < \infty.$$

Define the finite measures

$$\pi'_{m,n}(E) := \pi'(E \cap R_{m,n}), \quad \pi_{m,n}(E) := \pi(E \cap R_{m,n}).$$

For $A \in \mathcal{F}, B \in \mathcal{G}$,

$$\begin{aligned} \pi_{m,n}(A \times B) &= \pi((A \cap A_m) \times (B \cap B_n)) \\ &= \mu(A \cap A_m)\nu(B \cap B_n) \\ &= \pi'((A \cap A_m) \times (B \cap B_n)) \\ &= \pi'_{m,n}(A \times B). \end{aligned}$$

Thus, by the finite-case uniqueness theorem, $\pi_{m,n} = \pi'_{m,n}$ for every m, n . Hence, for every $E \in \mathcal{F} \otimes \mathcal{G}$,

$$\pi(E \cap R_{m,n}) = \pi'(E \cap R_{m,n}).$$

Since

$$X \times Y = \bigcup_{m=1}^{\infty} \bigcup_{n=1}^{\infty} R_{m,n}$$

is a countable disjoint union,

$$\begin{aligned} \pi(E) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi(E \cap R_{m,n}) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \pi'(E \cap R_{m,n}) = \pi'(E). \end{aligned}$$

Thus

$$\pi = \pi'.$$

□

Hence π is the desired product measure

$$\boxed{\mu \otimes \nu}.$$